

6. Data Flow and UML Diagrams

This section presents the analysis and design models of TriageAI. The models are derived from the Software Requirements Specification (SRS) Version 1.1, and every process, use case, and class traces to the functional (FR), nonfunctional (NFR), interface (IR), data (DR), security (SR), and business-rule (BR) requirements defined there. Section 6.1 models how data moves through the system using a system context diagram and leveled Data Flow Diagrams (DFDs). Section 6.2 models the system's behavior and structure using Unified Modeling Language (UML) diagrams: a use case diagram with detailed use case descriptions, activity diagrams, sequence diagrams, and a class/domain model. These models will serve as the basis for the architecture and detailed design in PD7.

6.1 Data Flow Diagrams

6.1.1 Notation and Conventions

The DFDs use the Gane and Sarson notation summarized below.

SYMBOL	MEANING
Shaded rectangle	External entity: a person, role, or outside system that sends data to or receives data from TriageAI.
Rounded rectangle with a number	Process: transforms incoming data flows into outgoing data flows. Level 1 processes are numbered 1.0 to 7.0; their Level 2 subprocesses are numbered 3.1, 3.2, and so on.
Open-ended rectangle labeled Dn	Data store: data at rest, such as a database table group. The same identifiers (D1 to D6) are used at every level.
Labeled arrow	Data flow: a named packet of data moving in the direction of the arrow. A label written as several lines on one arrow names several flows travelling together.
Dashed rounded rectangle	Used only in Level 2 diagrams: a neighbouring Level 1 process that exchanges data with the process being decomposed.

The following rules were applied: every process has at least one input and one output; data flows never connect an external entity directly to a data store or to another external entity; each data flow has a unique, meaningful name; and every child diagram is balanced with its parent, meaning that the flows entering and leaving a decomposed process are the same at both levels (Section 6.1.5).

6.1.2 System Context Diagram

The system context diagram (Figure 6.1) shows TriageAI as a single process (Process 0) and defines the system boundary. Six external entities interact with the system. Pet owners are not shown because, in the MVP, their symptom descriptions are relayed to the system by Intake Staff or

Veterinary Reviewers (SRS Section 2.3). Every flow in this diagram is traced to a Level 1 process in Table 6.1.5-A.

EXTERNAL ENTITY	DESCRIPTION
Intake Staff	Front-desk personnel who record new cases and monitor the Triage Queue and red-flag alerts.
Veterinary Reviewer	Licensed veterinarians and authorized technicians or nurses who review AI recommendations and make the final triage decision (Human-in-the-Loop).
Veterinary Domain Reviewer	A licensed veterinarian holding the knowledge base approval permission, who approves or rejects knowledge base entries.
Administrator	Manages accounts, drafts knowledge base entries, runs evaluations, and exports de-identified data.
LLM Service	The external transformer-based large language model (TBD-1) that performs entity extraction and recommendation generation on de-identified text.
Embedding Service	The external or locally hosted embedding model (TBD-2) that converts queries and knowledge base passages into vectors.

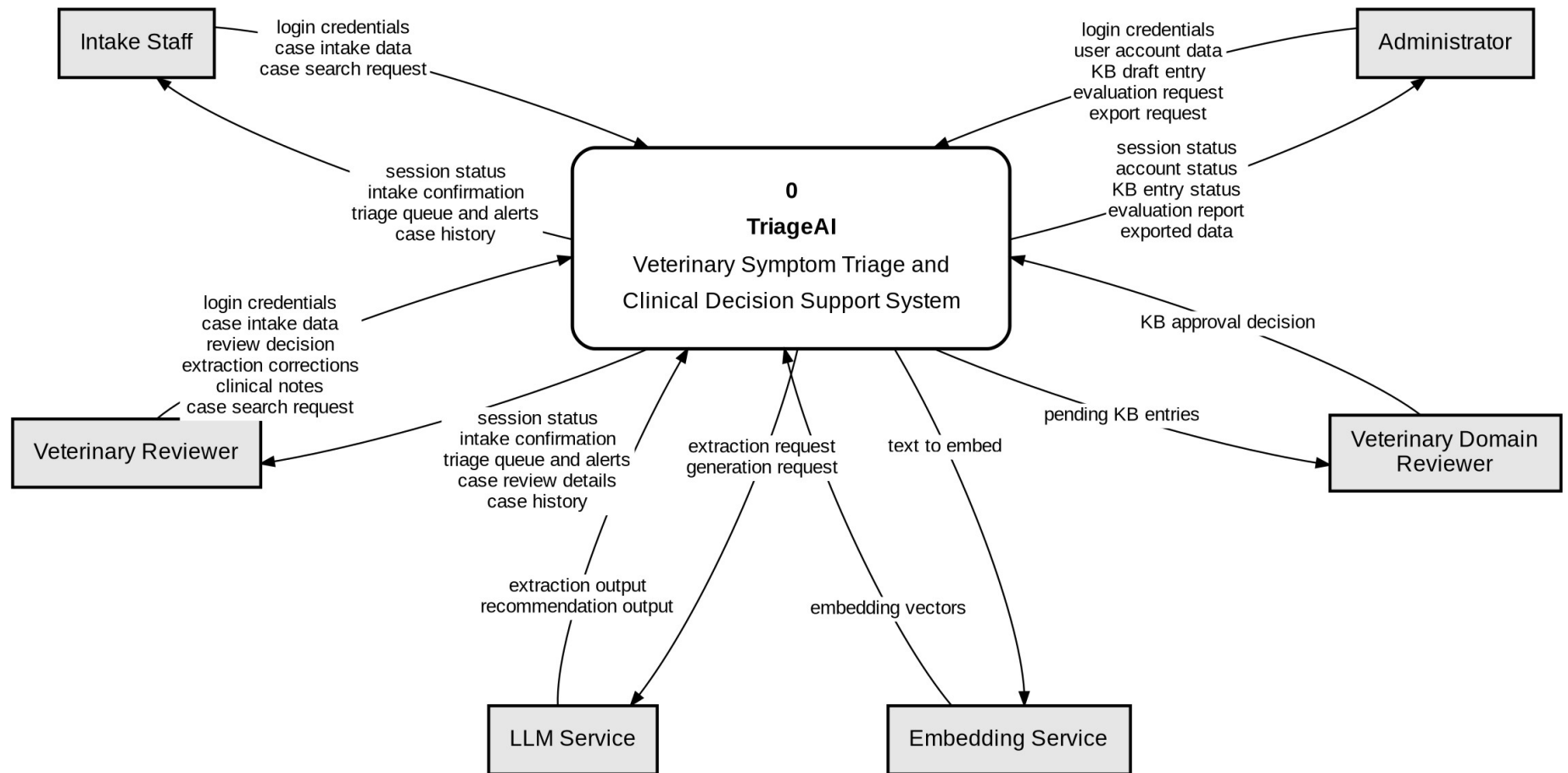


Figure 6.1: System Context Diagram of TriageAI

6.1.3 Level 1 Data Flow Diagram

The Level 1 DFD (Figure 6.2) decomposes Process 0 into seven major processes and six data stores. Processes 1.0 to 5.0 carry the live triage workflow, from login through intake, AI processing, and staff review; Process 6.0 maintains the curated knowledge base; and Process 7.0 supports the project evaluation. Every process that changes data sends an audit event to D4 Audit Log, which satisfies the audit-trail requirements (FR-41 to FR-43).

Process Descriptions

PROCESS	DESCRIPTION	RELATED REQUIREMENTS
1.0 Authenticate and Manage Users	Verifies login credentials, enforces logout and session rules, and lets the Administrator create, edit, and deactivate accounts.	FR-57 to FR-63; SR-01 to SR-05
2.0 Record Case Intake	Validates the intake form, stores the owner's description verbatim, creates the case with a unique ID, and starts the pipeline.	FR-01 to FR-08
3.0 Extract Clinical Entities	De-identifies the text, screens for red flags, obtains a structured extraction from the LLM, and validates and maps it to the supported complaint list.	FR-09 to FR-18; IR-13, IR-20
4.0 Generate VTL Recommendation	Retrieves the top-k knowledge base passages, obtains a VTL category with rationale and citations from the LLM, validates citations, and applies the red-flag safety floor.	FR-19 to FR-28; NFR-08, NFR-12
5.0 Manage Triage Review and Case Records	Builds the Triage Queue, presents cases for review, records staff decisions and notes, applies extraction corrections, and returns case histories.	FR-29 to FR-48; NFR-07
6.0 Manage Knowledge Base	Handles the Draft → Pending Review → Active → Retired workflow, veterinarian approval, chunking, embedding, and versioning.	FR-49 to FR-56; BR-04
7.0 Run Evaluation and Export	Imports labeled vignettes, runs them through the pipeline, computes metrics, and exports reports and de-identified data.	FR-47, FR-64 to FR-70; NFR-14

Data Store Descriptions

DATA STORE	CONTENTS	CLASSES (SECTION 6.2.5)
D1 Users and Roles	User accounts, roles, knowledge base approval permission, password hashes, and logout state.	User
D2 Case Records	Cases, signalment, verbatim owner descriptions, owner references (stored separately, DR-04), case status, staff decisions, and clinical notes.	Case, Signalment, OwnerDescription, OwnerReference, StaffDecision, ClinicalNote

DATA STORE	CONTENTS	CLASSES (SECTION 6.2.5)
D3 AI Outputs	Versioned extraction results, recommendations, and retrieved references, including evaluation outputs flagged as evaluation data.	ExtractionResult, TriageRecommendation, RetrievedReference
D4 Audit Log	Append-only audit entries for case and security events.	AuditEntry
D5 Knowledge Base	Entries, red-flag rules, the supported complaint list, chunks, embeddings (vector index), and immutable versions.	KnowledgeBaseEntry, RedFlagRule, PresentingComplaint, KnowledgeBaseChunk, KnowledgeBaseVersion
D6 Evaluation Data	Evaluation sets, vignettes, runs, and per-vignette results, stored separately from clinical cases (DR-06).	EvaluationSet, Vignette, EvaluationRun, EvaluationResult

6.1.4 Level 2 Data Flow Diagrams

Processes 3.0, 4.0, and 5.0 contain the core clinical decision-support logic and are decomposed further below. Processes 1.0, 2.0, 6.0, and 7.0 are considered functional primitives at this stage; their internal steps are shown in the activity diagrams in Section 6.2.3.

Process 3.0 Extract Clinical Entities (Figure 6.3). Process 3.1 removes owner identifiers before any text leaves the system (IR-20). Process 3.2 is a deterministic keyword screen that raises red flags immediately, independently of the LLM (FR-12, NFR-05). Process 3.3 sends the extraction request, and Process 3.4 validates the output against the schema, maps complaints to the supported list, and requests one retry before setting the case to Manual Triage Required (FR-15).

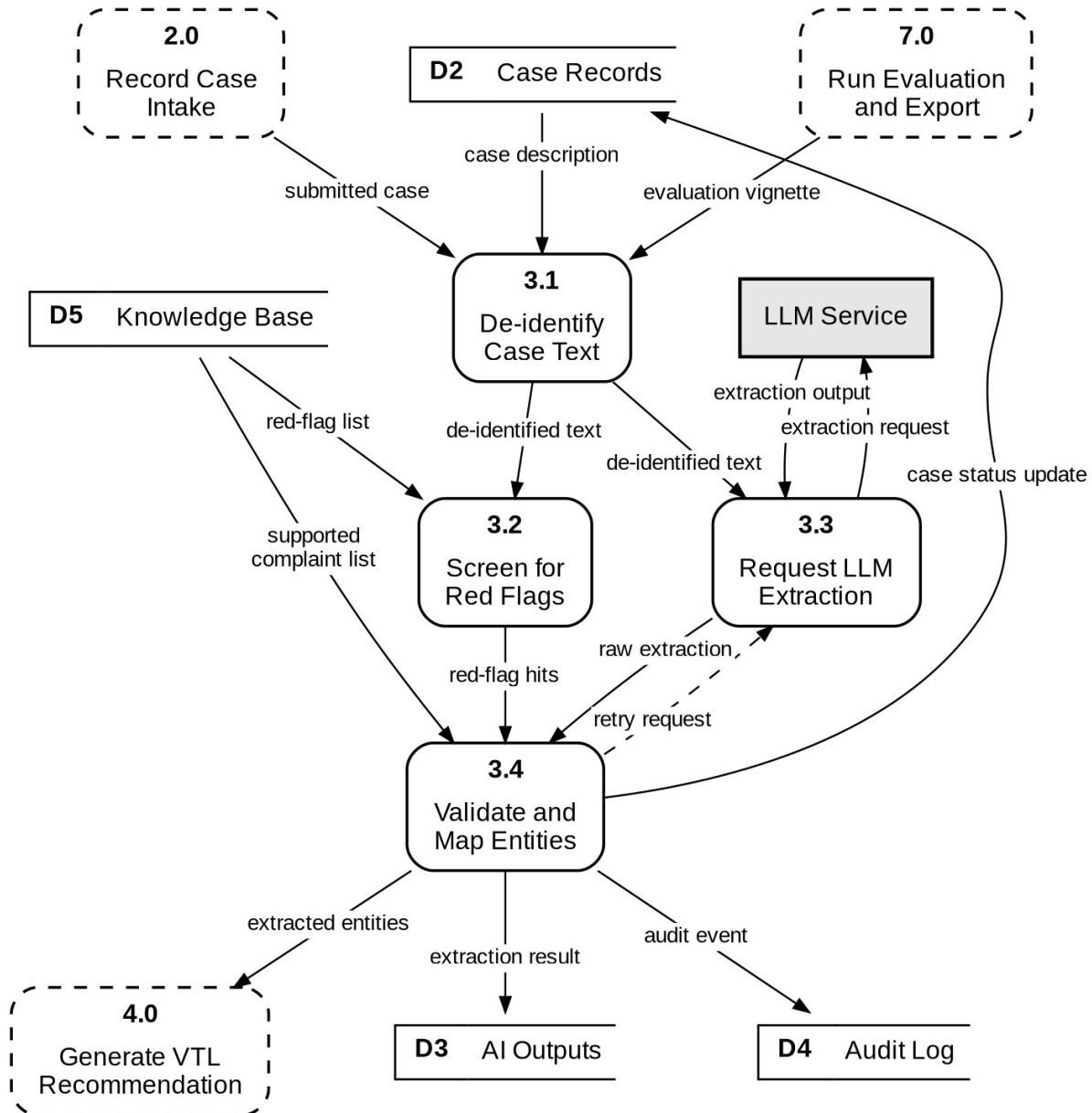


Figure 6.3: Level 2 DFD of Process 3.0 Extract Clinical Entities

Process 4.0 Generate VTL Recommendation (Figure 6.4). Process 4.1 turns the extracted entities, or a reviewer’s corrected entities, into a retrieval query and a case summary. Process 4.2 performs top-k similarity search over the active knowledge base version. Process 4.3 obtains the draft category, rationale, and citations from the LLM. Process 4.4 checks that every cited passage was actually retrieved (FR-22) and raises the category to the minimum defined for any detected red flag (FR-23) before the recommendation is stored.

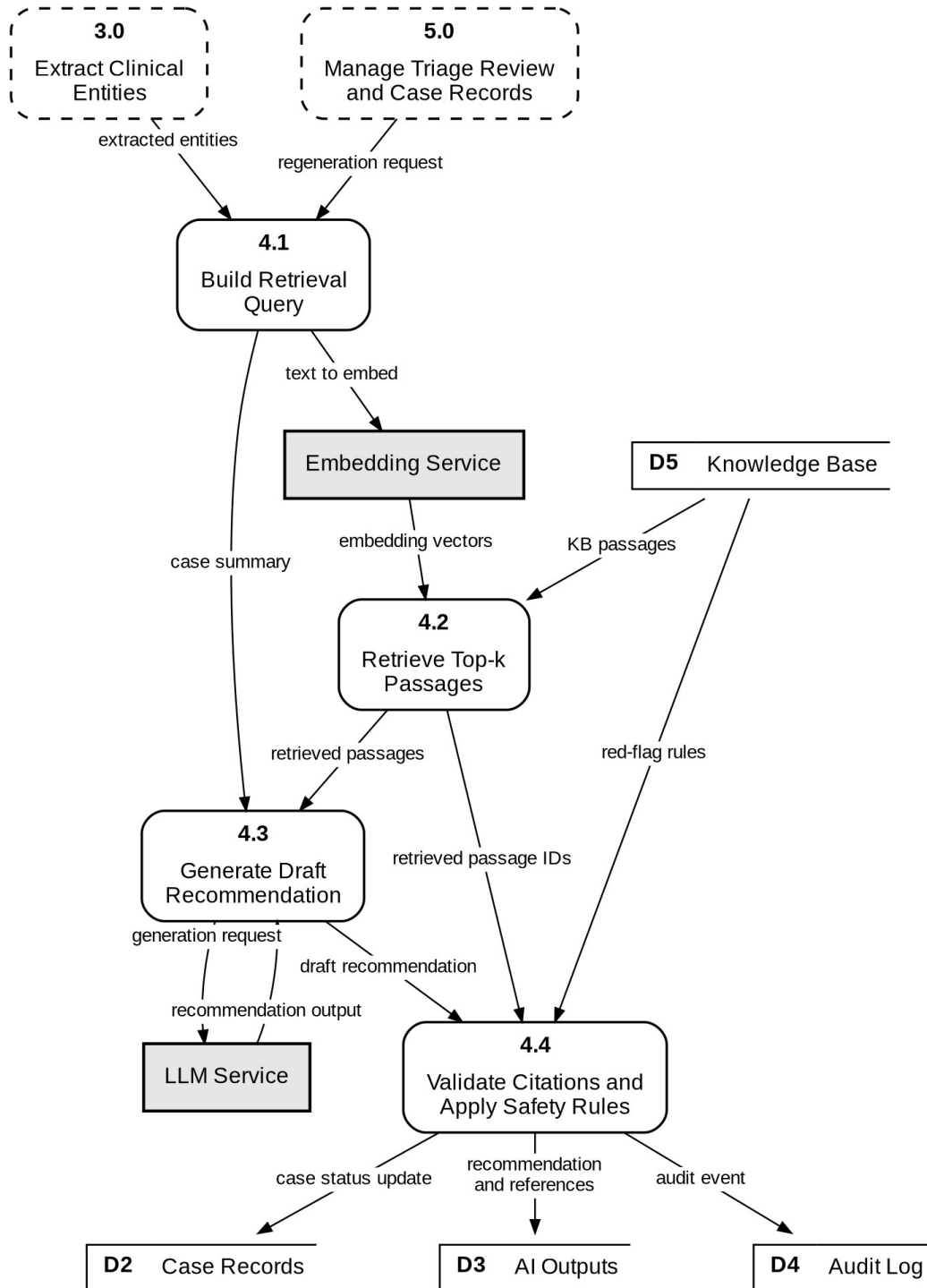


Figure 6.4: Level 2 DFD of Process 4.0 Generate VTL Recommendation

Process 5.0 Manage Triage Review and Case Records (Figure 6.5). Processes 5.1 and 5.2 read case and AI data to build the queue and the review screen; Process 5.3 records the final decision; Process 5.4 stores corrections and triggers regeneration through Process 4.0; and Process 5.5 answers case history searches using D2 and D4.

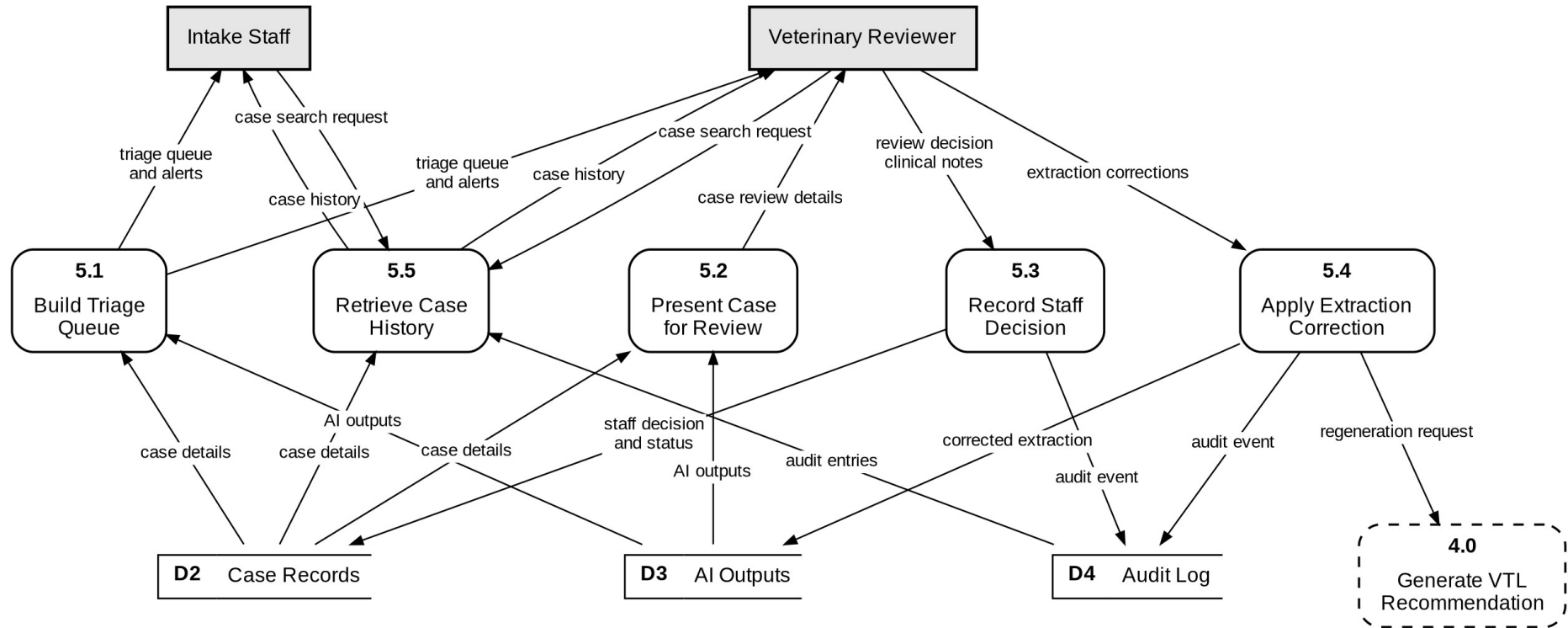


Figure 6.5: Level 2 DFD of Process 5.0 Manage Triage Review and Case Records

6.1.5 Balancing Verification

A DFD is balanced when the data flows entering and leaving a process are preserved when the process is decomposed. Table 6.1.5-A verifies the context diagram against the Level 1 DFD, and Tables 6.1.5-B to 6.1.5-D verify each Level 2 DFD against its parent process. A Level 1 flow may be split into more specific flows at Level 2 (marked “decomposed”); no flow is added or lost.

Table 6.1.5-A: Context Diagram to Level 1 DFD

EXTERNAL ENTITY	FLows INTO THE SYSTEM → LEVEL 1 PROCESS	FLows OUT OF THE SYSTEM ← LEVEL 1 PROCESS
Intake Staff	login credentials → 1.0 case intake data → 2.0 case search request → 5.0	session status ← 1.0 intake confirmation ← 2.0 triage queue and alerts ← 5.0 case history ← 5.0
Veterinary Reviewer	login credentials → 1.0 case intake data → 2.0 review decision, extraction corrections, clinical notes, case search request → 5.0	session status ← 1.0 intake confirmation ← 2.0 triage queue and alerts, case review details, case history ← 5.0
Administrator	login credentials, user account data → 1.0 KB draft entry → 6.0 evaluation request, export request → 7.0	session status, account status ← 1.0 KB entry status ← 6.0 evaluation report, exported data ← 7.0
Veterinary Domain Reviewer	KB approval decision → 6.0	pending KB entries ← 6.0
LLM Service	extraction output → 3.0 recommendation output → 4.0	extraction request ← 3.0 generation request ← 4.0
Embedding Service	embedding vectors → 4.0, 6.0	text to embed ← 4.0, 6.0

Table 6.1.5-B: Process 3.0 Extract Clinical Entities

LEVEL 1 FLOW	SOURCE → DESTINATION	LEVEL 2 PROCESS	BALANCED
submitted case	2.0 → 3.0	3.1	Yes
evaluation vignette	7.0 → 3.0	3.1	Yes
case description	D2 → 3.0	3.1	Yes

LEVEL 1 FLOW	SOURCE → DESTINATION	LEVEL 2 PROCESS	BALANCED
complaint and red-flag lists	D5 → 3.0	3.2 (red-flag list), 3.4 (supported complaint list)	Yes (decomposed)
extraction request	3.0 → LLM Service	3.3	Yes
extraction output	LLM Service → 3.0	3.3	Yes
extraction result	3.0 → D3	3.4	Yes
case status update	3.0 → D2	3.4	Yes
extracted entities	3.0 → 4.0	3.4	Yes
audit event	3.0 → D4	3.4	Yes

Internal flows (appear only at Level 2): de-identified text, red-flag hits, raw extraction, and retry request.

Table 6.1.5-C: Process 4.0 Generate VTL Recommendation

LEVEL 1 FLOW	SOURCE → DESTINATION	LEVEL 2 PROCESS	BALANCED
extracted entities	3.0 → 4.0	4.1	Yes
regeneration request	5.0 → 4.0	4.1	Yes
KB passages and red-flag rules	D5 → 4.0	4.2 (KB passages), 4.4 (red-flag rules)	Yes (decomposed)
text to embed	4.0 → Embedding Service	4.1	Yes
embedding vectors	Embedding Service → 4.0	4.2	Yes
generation request	4.0 → LLM Service	4.3	Yes
recommendation output	LLM Service → 4.0	4.3	Yes
recommendation and references	4.0 → D3	4.4	Yes
case status update	4.0 → D2	4.4	Yes
audit event	4.0 → D4	4.4	Yes

Internal flows (appear only at Level 2): case summary, retrieved passages, retrieved passage IDs, and draft recommendation.

Table 6.1.5-D: Process 5.0 Manage Triage Review and Case Records

LEVEL 1 FLOW	SOURCE → DESTINATION	LEVEL 2 PROCESS	BALANCED
case search request	Intake Staff, Veterinary Reviewer → 5.0	5.5	Yes
review decision, clinical notes	Veterinary Reviewer → 5.0	5.3	Yes
extraction corrections	Veterinary Reviewer → 5.0	5.4	Yes
triage queue and alerts	5.0 → Intake Staff, Veterinary Reviewer	5.1	Yes
case review details	5.0 → Veterinary Reviewer	5.2	Yes
case history	5.0 → Intake Staff, Veterinary Reviewer	5.5	Yes
case details	D2 → 5.0	5.1, 5.2, 5.5	Yes
AI outputs	D3 → 5.0	5.1, 5.2	Yes
audit entries	D4 → 5.0	5.5	Yes
staff decision and status	5.0 → D2	5.3	Yes
corrected extraction	5.0 → D3	5.4	Yes
regeneration request	5.0 → 4.0	5.4	Yes
audit event	5.0 → D4	5.3, 5.4	Yes

Internal flows (appear only at Level 2): none; each subprocess communicates only through the parent’s external flows and data stores.

6.2 UML Diagrams

6.2.1 Use Case Diagram

Figure 6.6 shows the sixteen use cases of TriageAI inside the system boundary. Primary actors appear on the left and right of the boundary; the LLM Service and the Embedding Service are secondary (supporting) actors. The Veterinary Domain Reviewer is a specialization of the Veterinary Reviewer, so it inherits all Veterinary Reviewer use cases and adds UC-14. UC-03 Process Triage Pipeline is included by every use case that sends text through the AI pipeline, and UC-06 to UC-09 extend UC-05 with the reviewer’s possible decisions.

6.2.2 Use Case Descriptions

The table below summarizes all use cases. Detailed descriptions follow for the use cases that carry the main triage workflow, the knowledge base approval, and the evaluation; the remaining use cases are simple create, read, update, or delete operations that are fully described by their summary and the related requirements.

ID	USE CASE	PRIMARY ACTOR(S)	BRIEF DESCRIPTION	REQUIREMENTS
UC-01	Log In	All users	Authenticate and start a session.	FR-57, FR-61, SR-02–SR-04
UC-02	Create Triage Case	Intake Staff, Veterinary Reviewer	Record species, signalment, and the owner's description.	FR-01–FR-08
UC-03	Process Triage Pipeline	System (LLM, Embedding)	Extract entities and generate a VTL recommendation.	FR-09–FR-28
UC-04	View Triage Queue	Intake Staff, Veterinary Reviewer	View open cases sorted by urgency with wait-time and red-flag alerts.	FR-29, FR-30, FR-39
UC-05	Review Case Recommendation	Veterinary Reviewer	Examine description, extraction, recommendation, and references.	FR-31, FR-36
UC-06	Confirm Recommendation	Veterinary Reviewer	Accept the recommended category as final.	FR-32, FR-35
UC-07	Adjust Urgency Category	Veterinary Reviewer	Change the category and record a reason.	FR-33, FR-35
UC-08	Perform Manual Triage	Veterinary Reviewer	Assign a category when the AI is not applicable.	FR-34, NFR-09
UC-09	Correct Extraction and Regenerate	Veterinary Reviewer	Fix extracted values and obtain a new recommendation.	FR-16, FR-27
UC-10	Close Case	Veterinary Reviewer	Mark the case as seen and remove it from the queue.	FR-38
UC-11	Search Case History	Intake Staff, Veterinary Reviewer	Search cases and view the audit timeline.	FR-44–FR-46
UC-12	Manage User Accounts	Administrator	Create, edit, deactivate accounts; assign roles.	FR-58–FR-60, FR-63
UC-13	Manage Knowledge Base Entries	Administrator	Draft, revise, submit, and retire entries.	FR-49–FR-51, FR-56
UC-14	Approve Knowledge Base Entry	Veterinary Domain Reviewer	Approve or reject entries; activate and index them.	FR-52–FR-54
UC-15	Run System Evaluation	Administrator	Import vignettes, run the pipeline, and compute metrics.	FR-64–FR-70

ID	USE CASE	PRIMARY ACTOR(S)	BRIEF DESCRIPTION	REQUIREMENTS
UC-16	Export De-identified Data	Administrator	Export case, audit, and evaluation data as CSV/JSON.	FR-47, FR-69, BR-08

UC-01: Log In

Primary Actor	Intake Staff, Veterinary Reviewer, or Administrator
Secondary Actors	None
Description	A registered user authenticates to gain access to the functions permitted for the user's role.
Trigger	The user opens TriageAI in a browser.
Preconditions	The user has an active account.
Postconditions	An authenticated session exists and the Triage Queue (or Administrator Console) is displayed.
Main Success Scenario	1. The user enters a username or e-mail address and a password. 2. The system verifies the credentials against the stored password hash. 3. The system resets the failed-attempt counter and creates a session. 4. The system displays the home screen for the user's role.
Alternative Flows	2a. First login with a temporary password: the system requires a new password before continuing (FR-61).
Exceptions	2b. Invalid credentials: the system shows a generic error and increments the failed-attempt counter. 2c. Five consecutive failures: the account is locked for 15 minutes (SR-03).
Business Rules	SR-02, SR-04
Related Requirements	FR-57, FR-61, FR-62, SR-01–SR-05, SR-12
Priority	High

UC-02: Create Triage Case

Primary Actor	Intake Staff or Veterinary Reviewer
Secondary Actors	LLM Service, Embedding Service (through UC-03)

Description	The user records a new case from a walk-in, phone, or online message so that it can be prioritized.
Trigger	An owner reports a pet's condition to the clinic.
Preconditions	The user is logged in (UC-01).
Postconditions	A case exists with status Submitted and the triage pipeline (UC-03) has started; the case appears in the Triage Queue.
Main Success Scenario	1. The user selects New Case. 2. The system displays the Case Intake Form. 3. The user selects the species and enters optional signalment, the intake channel, and the owner's description. 4. The user submits the form. 5. The system validates the input, stores the description verbatim, and creates the case with a unique ID and timestamp. 6. The system includes UC-03 Process Triage Pipeline and shows a processing indicator. 7. The system adds the case to the Triage Queue.
Alternative Flows	3a. The user enters an optional owner reference: the system stores it separately and excludes it from AI processing (FR-07).
Exceptions	5a. A required field is missing or the description is outside 20–2,000 characters: the system highlights the fields and keeps the entered data (FR-04). 5b. The species is not a dog or cat: the system does not start UC-03 and informs the user to triage manually (FR-02).
Business Rules	BR-06
Related Requirements	FR-01–FR-08
Priority	High

UC-03: Process Triage Pipeline

Primary Actor	System (initiated by UC-02, UC-09, or UC-15)
Secondary Actors	LLM Service, Embedding Service
Description	The system converts the owner's description into structured clinical information and a VTL recommendation grounded in the curated knowledge base.
Trigger	A case is submitted, a correction is saved, or an evaluation vignette is processed.

Preconditions	The case text is stored; an active knowledge base version exists.
Postconditions	An extraction result and a recommendation with references are stored and the case status is Awaiting Review, or the status is Manual Triage Required.
Main Success Scenario	1. The system removes owner identifiers from the text. 2. The system runs the deterministic red-flag pre-screen and raises any alert. 3. The system sends the extraction request to the LLM Service. 4. The system validates the extraction against the schema and maps complaints to the supported list. 5. The system builds a query, obtains its embedding, and retrieves the top-k passages. 6. The system sends the generation request with the extracted entities and passages to the LLM Service. 7. The system validates the citations and applies the red-flag safety floor. 8. The system stores the recommendation, its references, and the configuration used, and sets the status to Awaiting Review.
Alternative Flows	4a. The complaint is not on the supported list: it is labeled Other/Unsupported and manual triage is recommended (FR-10). 7a. A citation does not match a retrieved passage or confidence is low: the recommendation is labeled low confidence (FR-22, FR-24).
Exceptions	3a/6a. The LLM Service times out or returns invalid output after one retry: the status becomes Manual Triage Required and users are notified (FR-15, IR-19, NFR-09).
Business Rules	FR-25 (choose the more urgent of two adjacent categories)
Related Requirements	FR-09–FR-28, IR-12–IR-14, IR-19, IR-20, NFR-01, NFR-05
Priority	High

UC-05: Review Case Recommendation

Primary Actor	Veterinary Reviewer
Secondary Actors	None
Description	The reviewer examines the AI output for a case before deciding on its urgency.
Trigger	The reviewer selects a case in the Triage Queue.
Preconditions	The case status is Awaiting Review or Manual Triage Required.

Postconditions	The reviewer has the information needed to perform UC-06, UC-07, UC-08, or UC-09.
Main Success Scenario	1. The reviewer opens the case from the Triage Queue. 2. The system displays the original description with highlighted evidence, the extracted information, the recommended category and target waiting time, the rationale, the confidence indicator, and the references. 3. The reviewer compares the extracted information with the description and reads the references. 4. The reviewer selects one of the extending use cases.
Alternative Flows	2a. The case has a red-flag alert: an alert banner is displayed at the top of the screen.
Exceptions	1a. The user is Intake Staff: the case is shown read-only and decision actions are unavailable (FR-35).
Business Rules	BR-01, BR-02
Related Requirements	FR-31, FR-35, FR-36, IR-03, IR-04, IR-06
Priority	High

UC-06 / UC-07: Confirm Recommendation / Adjust Urgency Category

Primary Actor	Veterinary Reviewer
Secondary Actors	None
Description	The reviewer finalizes the urgency category by accepting (UC-06) or changing (UC-07) the AI recommendation.
Trigger	The reviewer clicks Confirm or Adjust Category on the Case Review screen.
Preconditions	UC-05 is in progress.
Postconditions	A StaffDecision is stored, the status is Confirmed or Adjusted, an audit entry is written, and the queue is re-sorted.
Main Success Scenario	1. The reviewer clicks Confirm (UC-06) or selects a different category and a reason (UC-07). 2. The system verifies that the user has the Veterinary Reviewer role. 3. The system stores the decision with the reviewer, timestamp, and, for UC-07, the reason and direction (up- or down-triage). 4. The system writes an audit entry with the previous and new values. 5. The system displays the final category and re-sorts the Triage Queue.
Alternative Flows	None

Exceptions	1a. UC-07 without a reason: the system requires a reason before saving (FR-33). 2a. The user lacks the role: the system rejects the request (HTTP 403).
Business Rules	BR-01, BR-03; a finalized decision can only be amended (FR-48)
Related Requirements	FR-32, FR-33, FR-35, FR-42, FR-48, NFR-07, NFR-11
Priority	High

UC-08: Perform Manual Triage

Primary Actor	Veterinary Reviewer
Secondary Actors	None
Description	The reviewer assigns a category without relying on the AI when the AI output is unavailable or not applicable.
Trigger	The case status is Manual Triage Required, or the reviewer judges the AI output not applicable.
Preconditions	UC-05 is in progress.
Postconditions	The case has a staff-assigned category with status Manually Triaged.
Main Success Scenario	1. The reviewer selects Mark for Manual Triage. 2. The system displays the VTL reference with the five categories and target waiting times. 3. The reviewer selects a category and enters a note. 4. The system stores the decision and writes an audit entry.
Alternative Flows	None
Exceptions	None
Business Rules	BR-06
Related Requirements	FR-34, NFR-09, NFR-13
Priority	High

UC-09: Correct Extraction and Regenerate

Primary Actor	Veterinary Reviewer
Secondary Actors	LLM Service, Embedding Service (through UC-03)

Description	The reviewer corrects extracted values and obtains a new recommendation based on the corrected information.
Trigger	The reviewer finds an incorrect or missing extracted value.
Preconditions	UC-05 is in progress.
Postconditions	A corrected extraction and a new recommendation version are stored; earlier versions are retained.
Main Success Scenario	1. The reviewer edits one or more extracted fields and clicks Regenerate. 2. The system stores the corrected values alongside the original AI values. 3. The system includes UC-03 starting from retrieval with the corrected entities. 4. The system displays the new recommendation and returns to UC-05.
Alternative Flows	None
Exceptions	3a. The pipeline fails: the previous recommendation remains visible and the case is flagged for manual triage.
Business Rules	None
Related Requirements	FR-16, FR-27, FR-26
Priority	High

UC-14: Approve Knowledge Base Entry

Primary Actor	Veterinary Domain Reviewer
Secondary Actors	Embedding Service
Description	A licensed veterinarian verifies a drafted entry against its source and approves or rejects it.
Trigger	An entry has status Pending Review.
Preconditions	The reviewer is logged in and holds the knowledge base approval permission.
Postconditions	Approved: the entry is Active, embedded, indexed, and included in a new immutable KB version. Rejected: the entry returns to Draft with comments.

Main Success Scenario	1. The reviewer opens the list of pending entries. 2. The reviewer compares the entry with its cited source. 3. The reviewer approves the entry. 4. The system chunks the content, obtains embeddings, and updates the vector index. 5. The system sets the entry to Active, publishes a new KB version, and writes an audit entry.
Alternative Flows	3a. The reviewer rejects the entry with comments: the system returns it to Draft for revision (UC-13).
Exceptions	3b. The user lacks the approval permission: the action is refused (HTTP 403).
Business Rules	BR-04
Related Requirements	FR-50–FR-55, NFR-08
Priority	High

UC-15: Run System Evaluation

Primary Actor	Administrator
Secondary Actors	LLM Service, Embedding Service (through UC-03)
Description	The Administrator runs the labeled vignette set through the pipeline and obtains the evaluation metrics.
Trigger	The team schedules an evaluation.
Preconditions	A validated evaluation set of 50–100 labeled vignettes exists.
Postconditions	An EvaluationRun with metrics and per-vignette results is stored and can be exported (UC-16).
Main Success Scenario	1. The Administrator starts an evaluation run for a selected set. 2. The system records the configuration (model, prompt version, KB version). 3. For each vignette, the system includes UC-03 and records the outputs and stage latencies. 4. The system computes classification, retrieval, and latency metrics and lists under-triaged vignettes. 5. The system displays the evaluation report.
Alternative Flows	None

Exceptions	3a. An AI service fails for a vignette: the vignette is recorded as failed and the run continues.
Business Rules	Evaluation data is stored separately from clinical cases (DR-06)
Related Requirements	FR-64–FR-70, NFR-14, NFR-16, NFR-17
Priority	Medium

6.2.3 Activity Diagrams

The activity diagrams model the four key workflows using swimlanes to show which actor or system performs each action. Decision nodes correspond to the alternative and exception flows of the use cases above.

AD-1: Case Intake and Recommendation Generation (Figure 6.7) covers UC-02 and UC-03, from form entry to the recommendation or the fallback to manual triage.

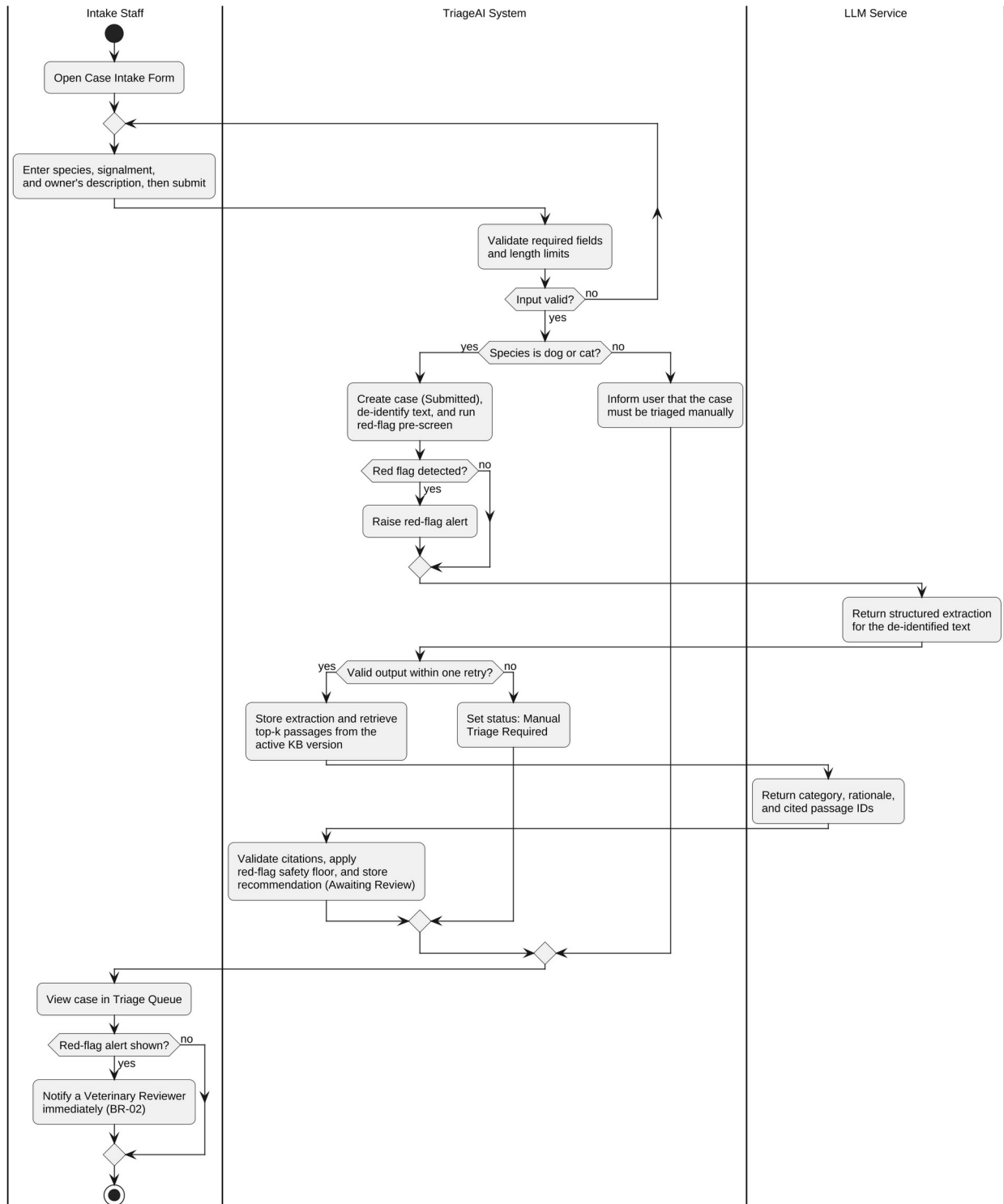


Figure 6.7: AD-1 Case Intake and Recommendation Generation

AD-2: Human-in-the-Loop Review and Decision (Figure 6.8) covers UC-05 to UC-10. The loop allows the reviewer to correct the extraction and regenerate as many times as needed before deciding.

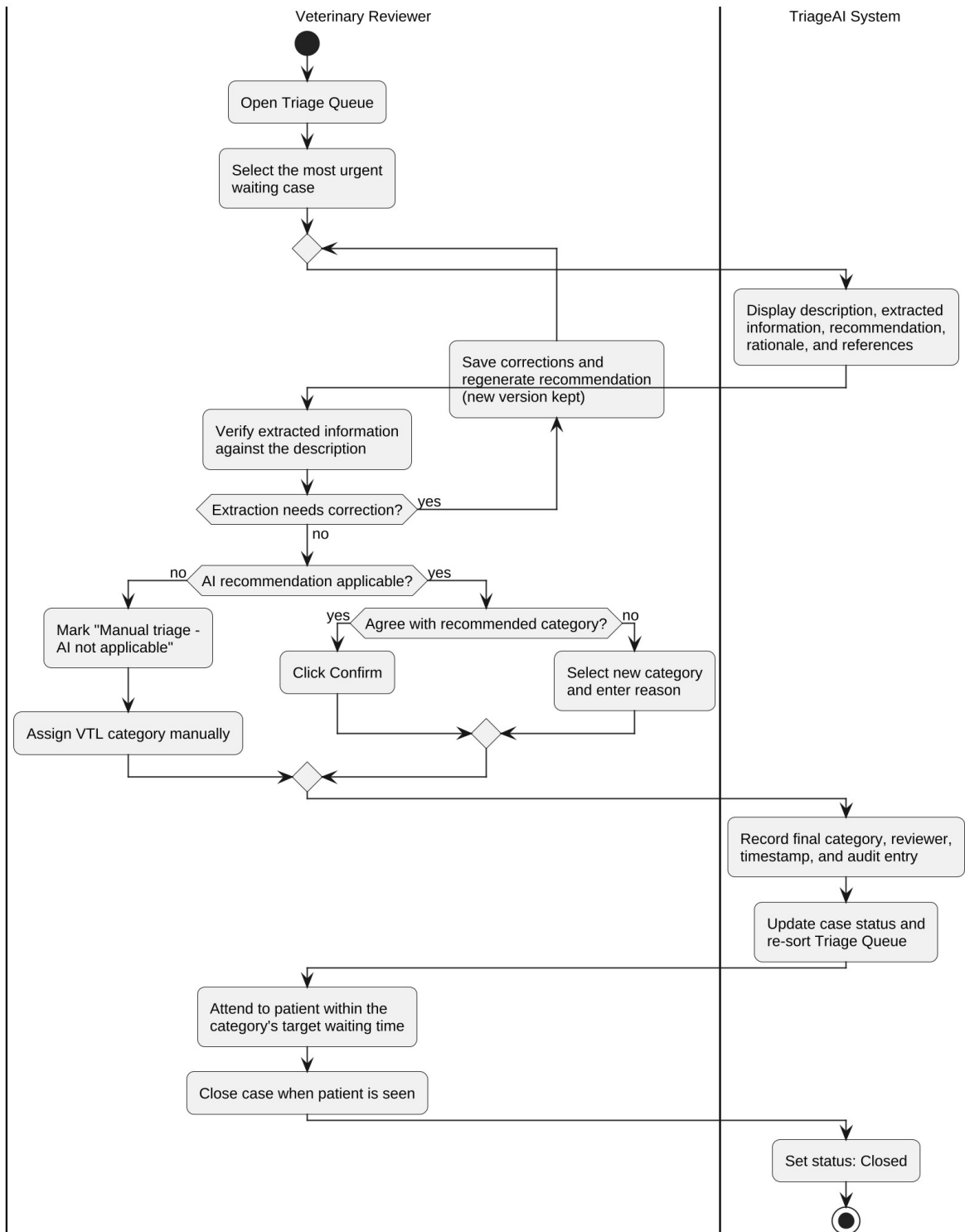


Figure 6.8: AD-2 Human-in-the-Loop Review and Decision

AD-3: Knowledge Base Curation and Approval (Figure 6.9) covers UC-13 and UC-14. An entry is used for retrieval only after a licensed veterinarian approves it.

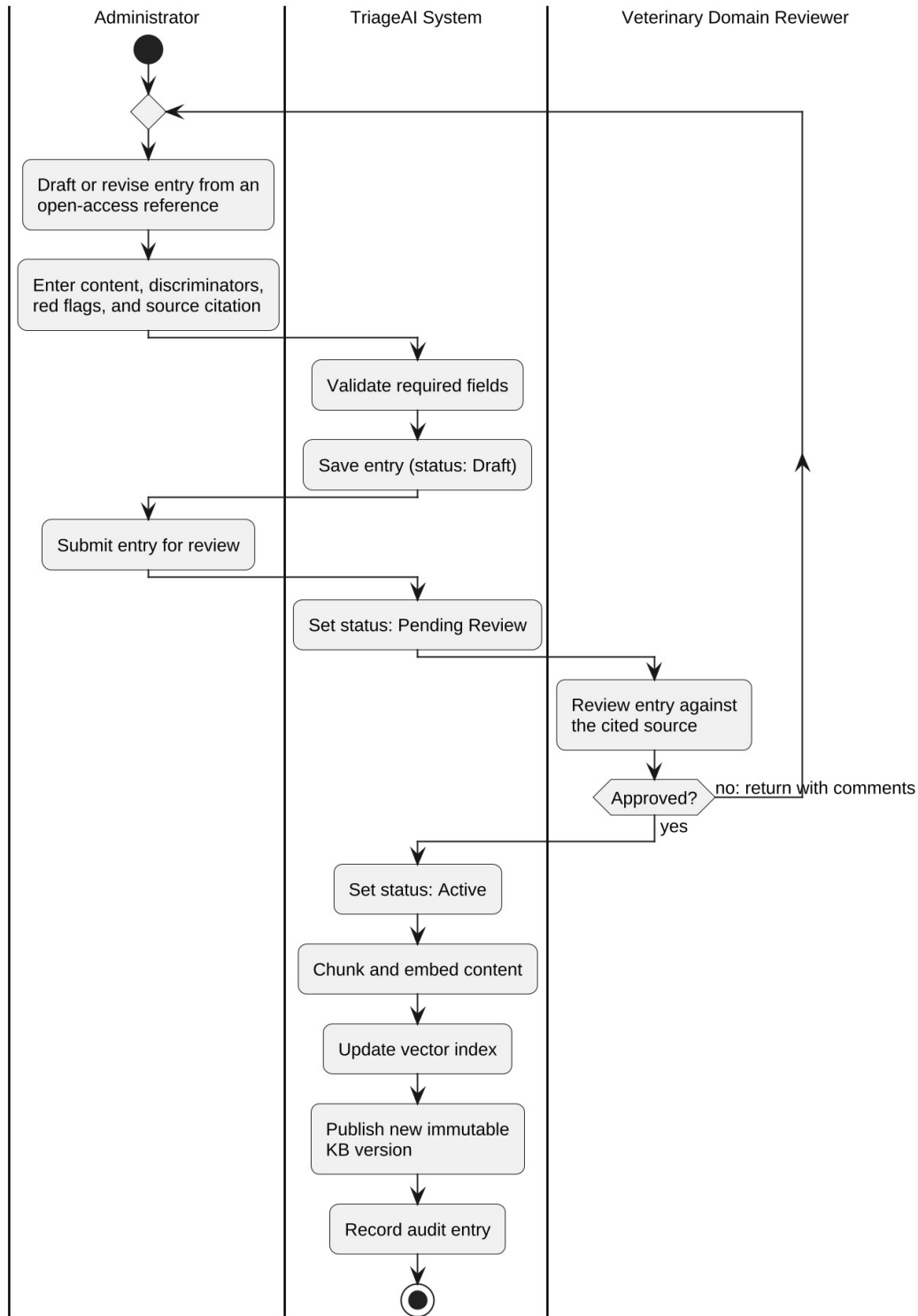


Figure 6.9: AD-3 Knowledge Base Curation and Approval

AD-4: System Evaluation Run (Figure 6.10) covers UC-15 and UC-16. Metric computation is shown as parallel activities because the three metric groups are independent.

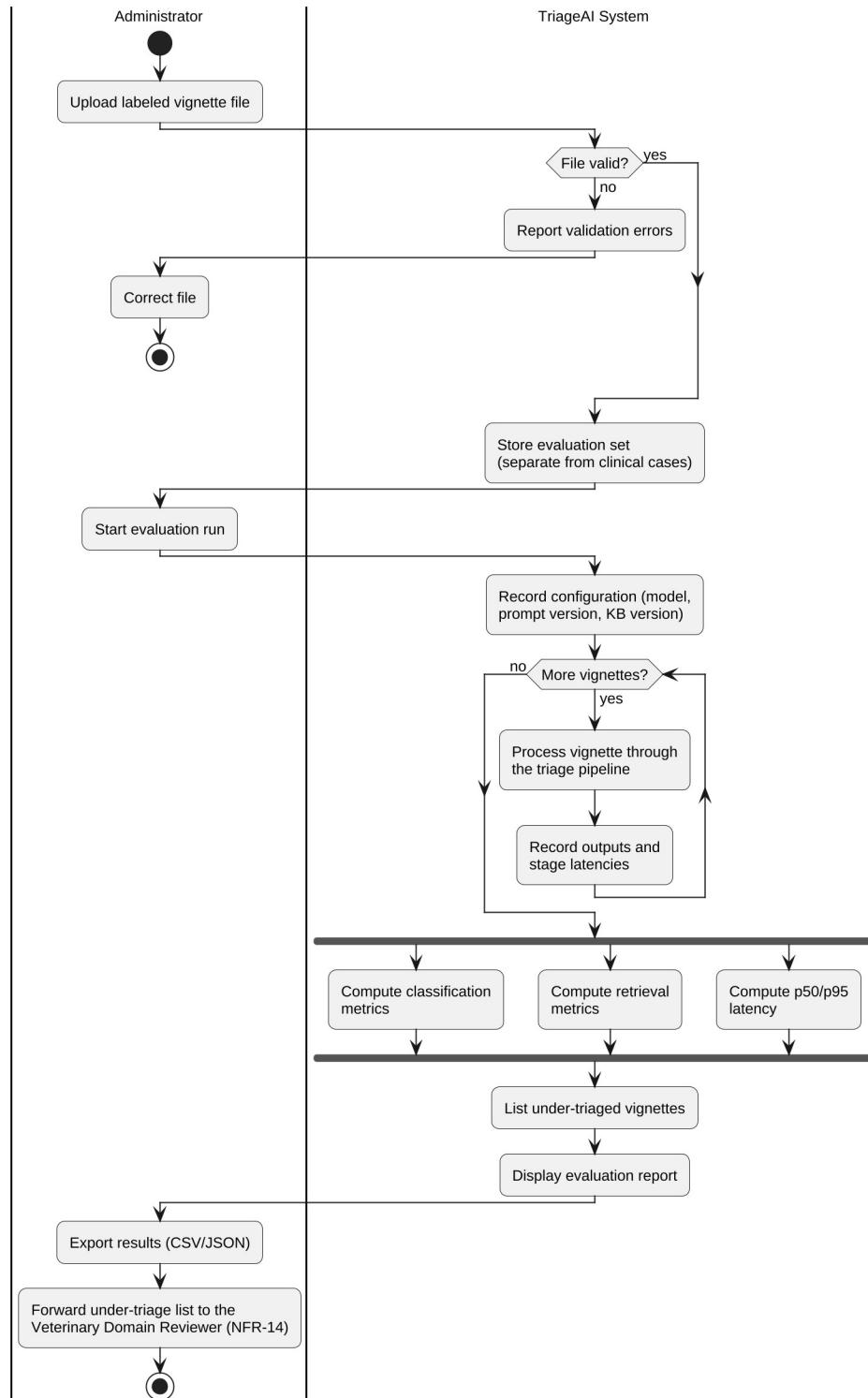


Figure 6.10: AD-4 System Evaluation Run

6.2.4 Sequence Diagrams

The sequence diagrams show the messages exchanged between the actors and the software components identified in the SRS (Section 2.1) for the four major interactions. Messages are numbered in order; alt, opt, and loop fragments show conditional and repeated behavior. The REST endpoints shown are provisional and will be finalized in the API design in PD7.

SD-1 corresponds to UC-02 and UC-03. The API responds to the browser immediately (message 5) and the pipeline continues asynchronously; the client polls for the result (IR-22).

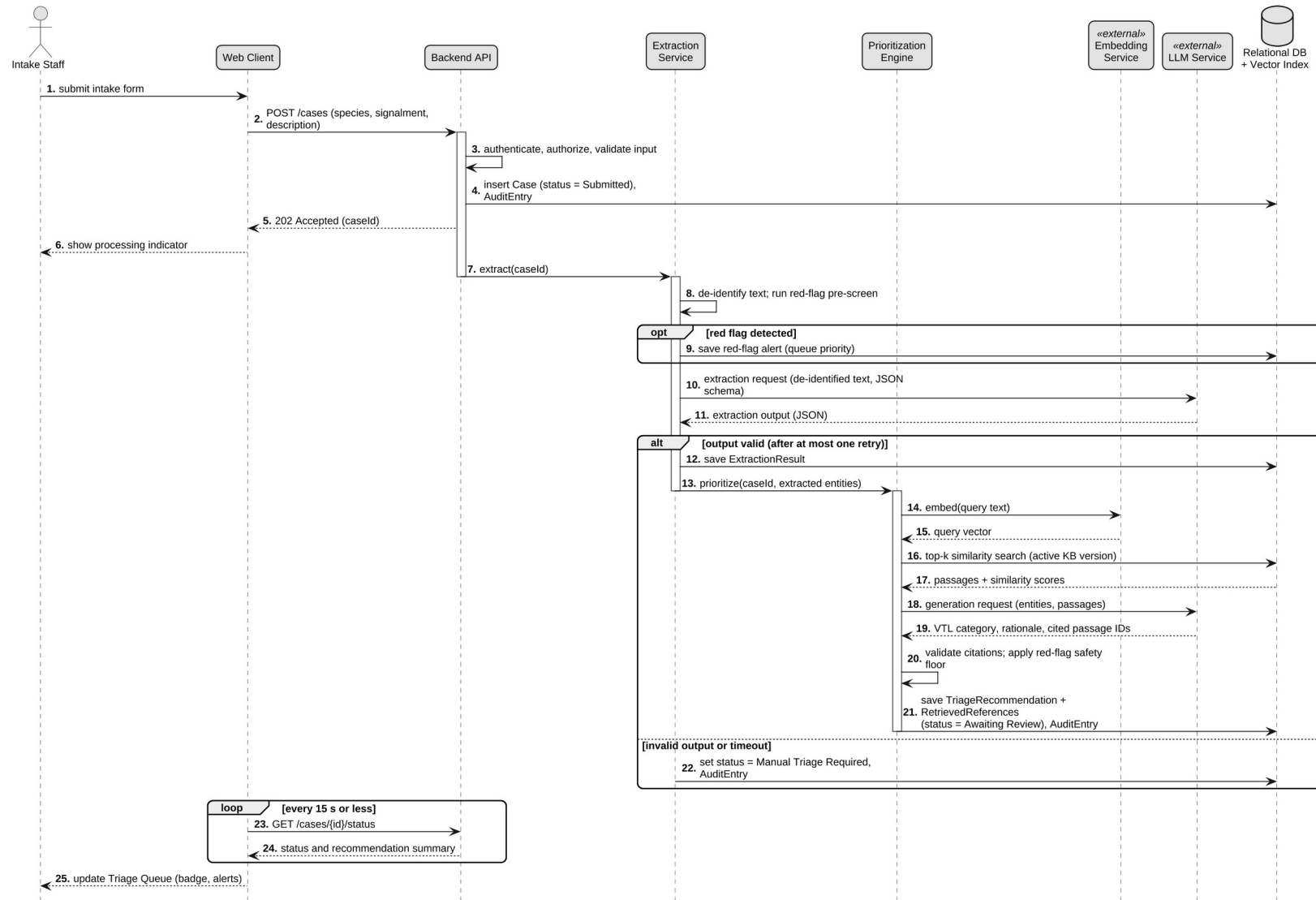


Figure 6.11: SD-1 Submit Case and Generate Recommendation

SD-2 corresponds to UC-06 and UC-07. Role checking is performed on the server for every decision request (SR-05), not only in the user interface.

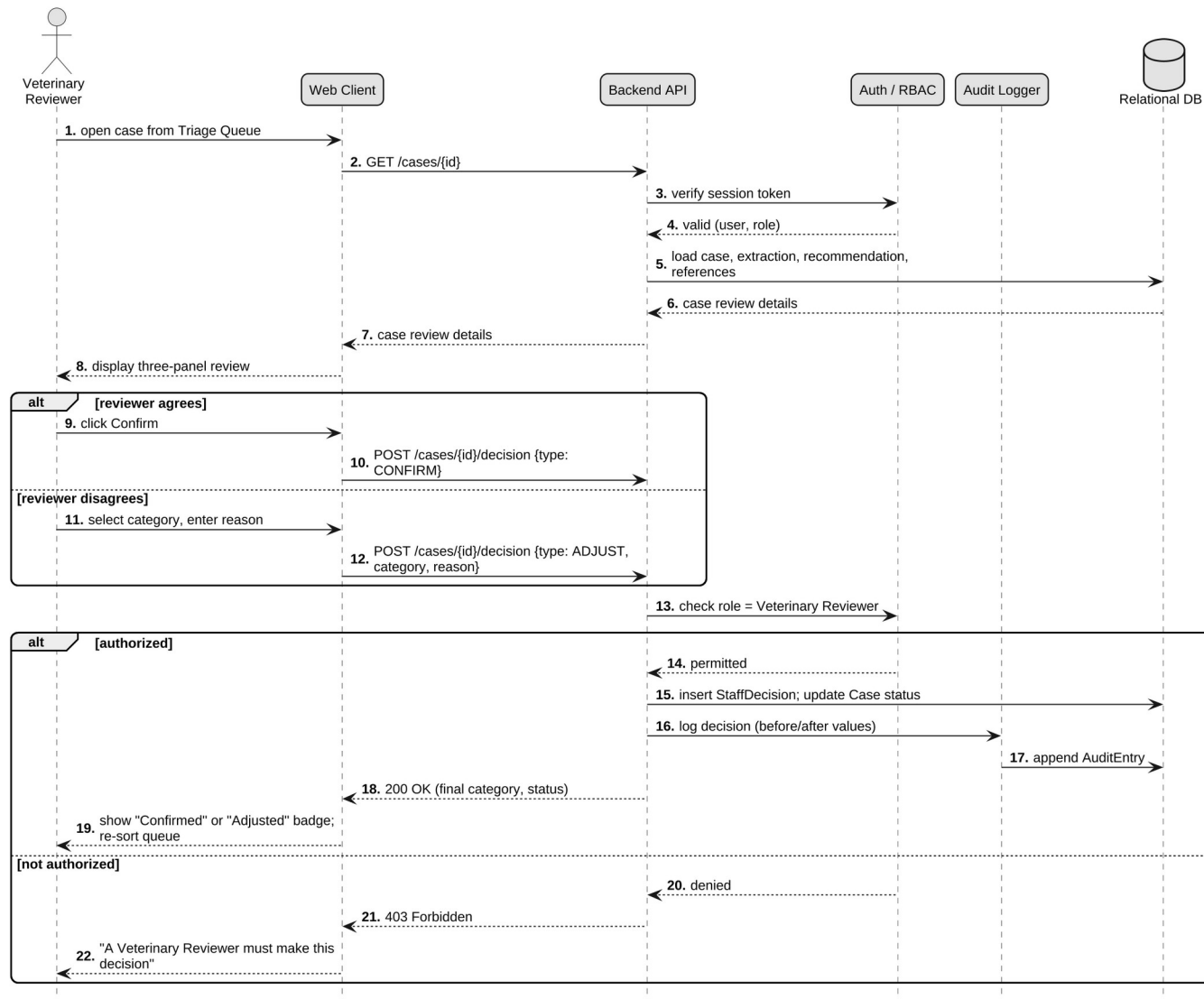


Figure 6.12: SD-2 Confirm or Adjust a Recommendation

SD-3 corresponds to UC-13 and UC-14. A new immutable knowledge base version is published only after all chunks have been embedded and indexed.

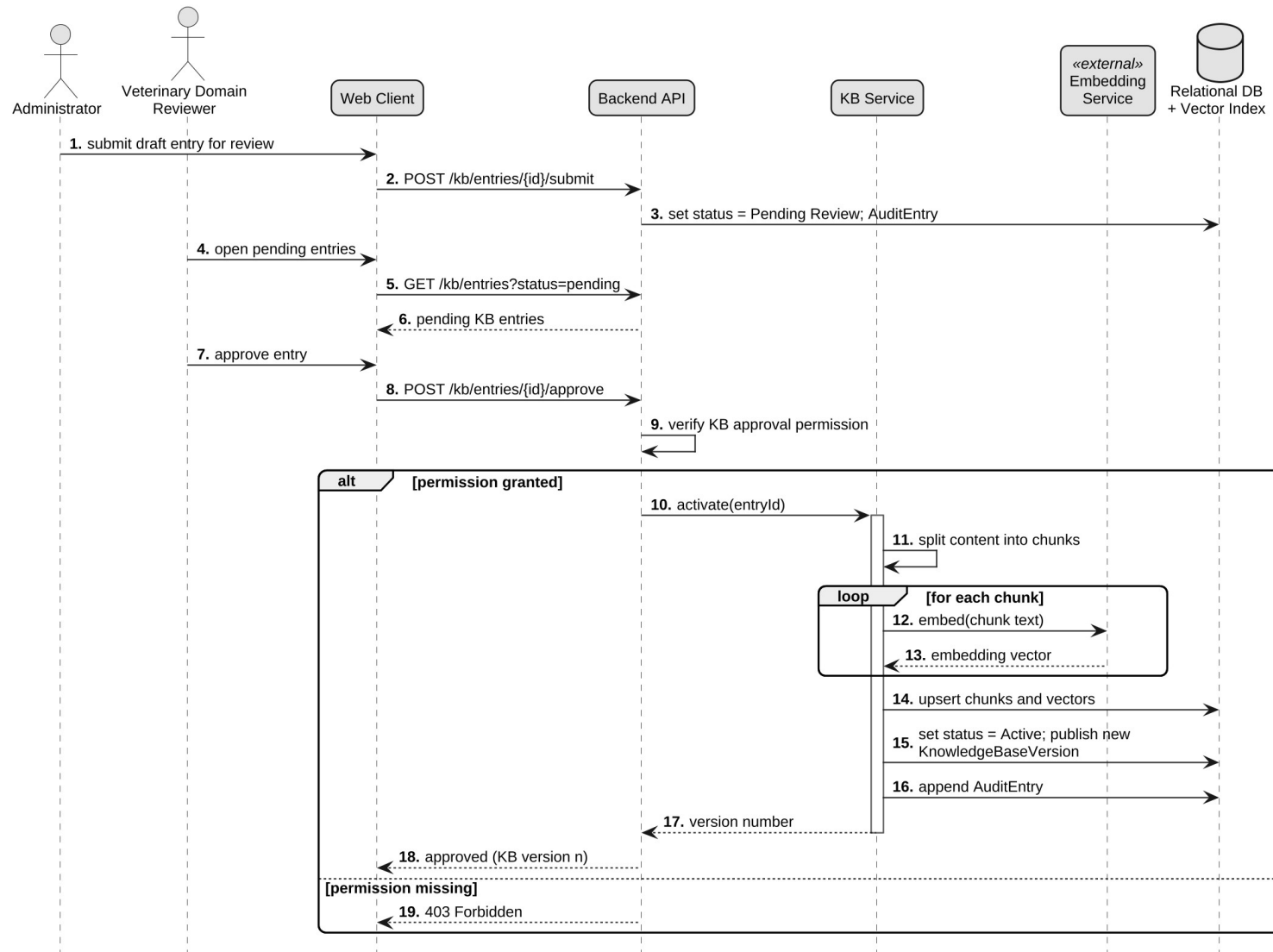


Figure 6.13: SD-3 Approve Knowledge Base Entry and Re-index

SD-4 corresponds to UC-09. The corrected extraction and the new recommendation are stored as new versions so that the original AI output remains auditable.

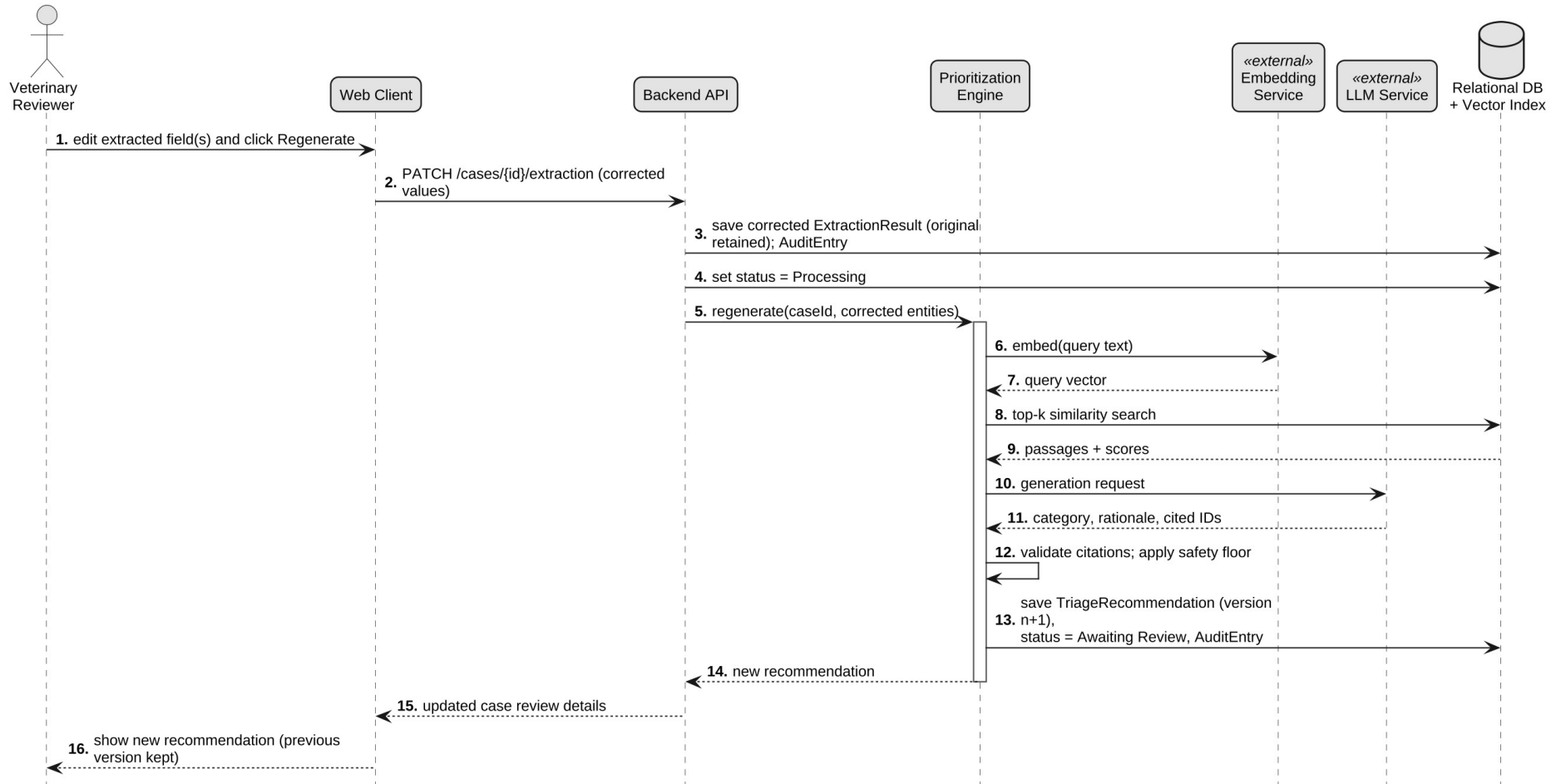


Figure 6.14: SD-4 Correct Extraction and Regenerate Recommendation

6.2.5 Class/Domain Model

The domain model identifies the persistent classes of TriageAI, their key attributes and operations, and the relationships between them. It refines the core entities listed in DR-01. For readability, it is divided into two diagrams: (a) accounts, triage cases, and AI outputs; and (b) the knowledge base, evaluation, and enumerations. Classes shown with “(see model a/b)” are defined in the other diagram. Filled diamonds denote composition (the part cannot exist without the whole), hollow diamonds denote aggregation, and plain arrows denote directed associations.

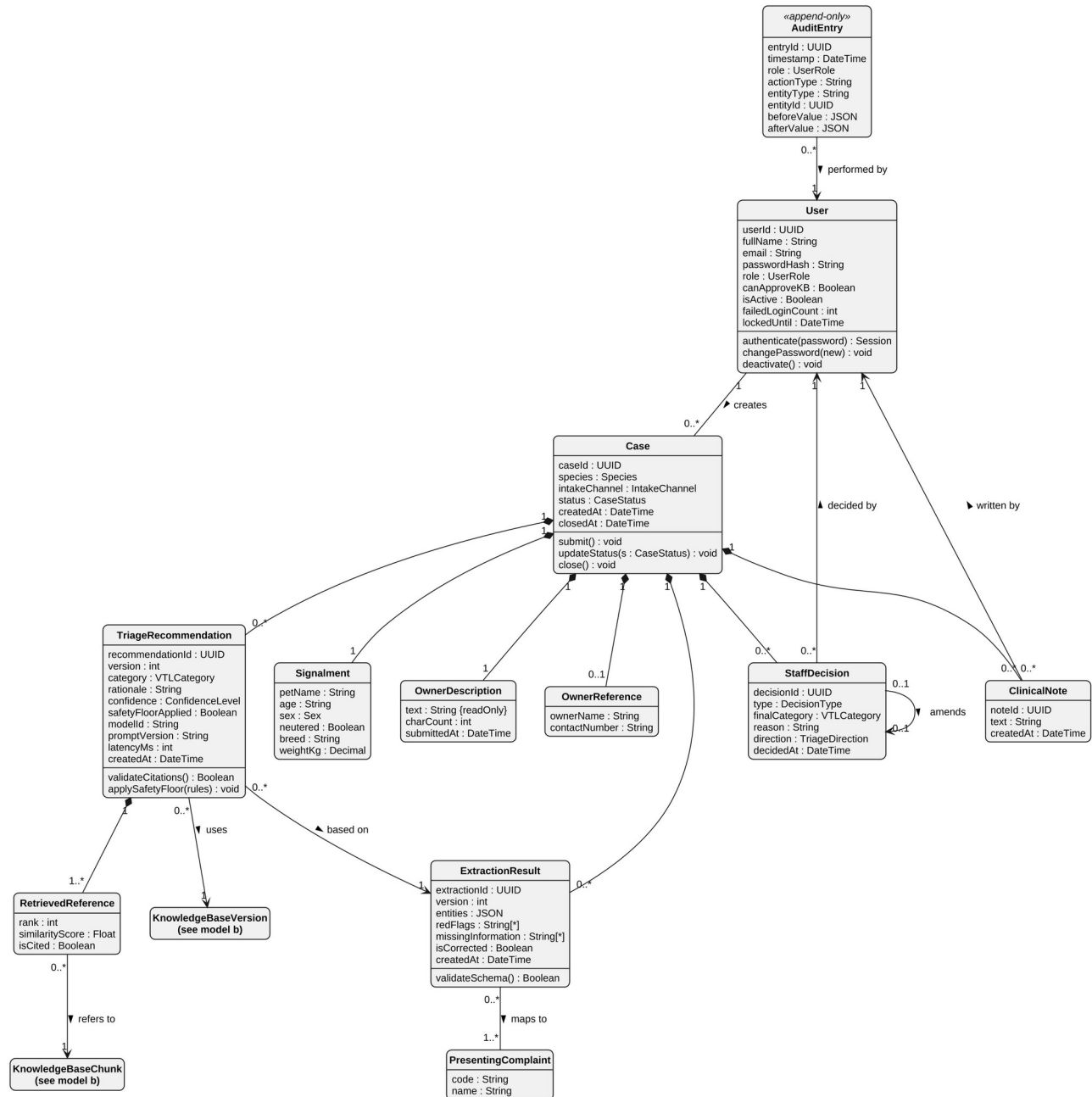


Figure 6.15: Domain Model (a) – Accounts, Triage Cases, and AI Outputs

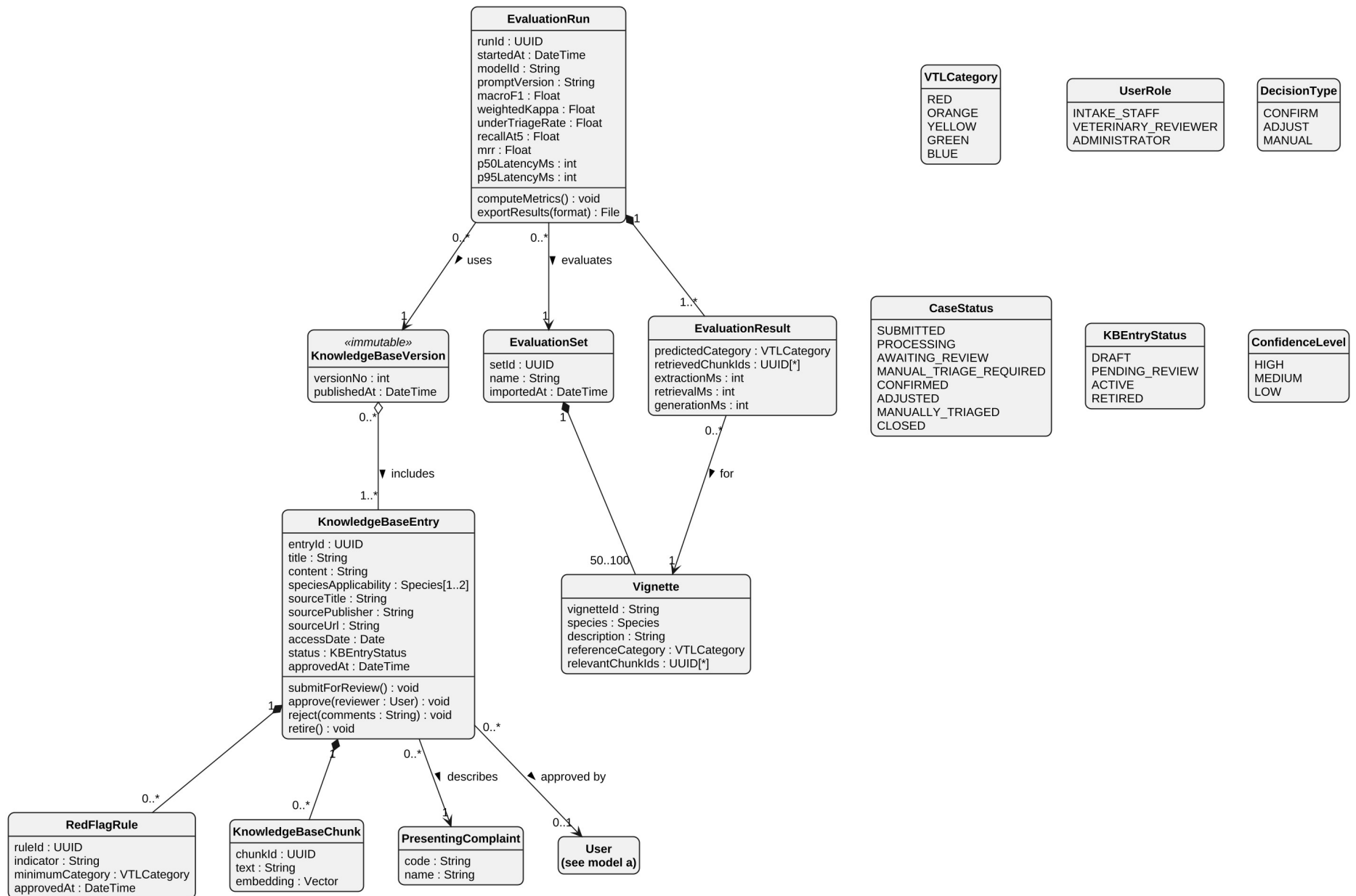


Figure 6.16: Domain Model (b) – Knowledge Base, Evaluation, and Enumerations

Class Dictionary

CLASS	RESPONSIBILITY	KEY RELATIONSHIPS AND CONSTRAINTS
User	A registered staff account with one role and an optional KB approval permission.	Creates Cases; decides StaffDecisions; performs AuditEntries.
Case	One triage encounter for a dog or cat; its status follows the state model in SRS Appendix B.	Composed of one Signalment and one OwnerDescription; has versioned ExtractionResults and TriageRecommendations.
Signalment	Pet name, age, sex and neuter status, breed, and weight.	Part of exactly one Case.
OwnerDescription	The owner's description stored verbatim.	Read-only after submission (FR-05).
OwnerReference	Optional owner name and contact number.	Stored separately and never sent to external services (DR-04, IR-20).
ExtractionResult	A versioned structured extraction, including red flags and missing information.	Maps to one or more PresentingComplaints; corrected versions keep the original (FR-16).
TriageRecommendation	A versioned VTL category with rationale, confidence, and the configuration used.	Based on one ExtractionResult; uses one KnowledgeBaseVersion; has one or more RetrievedReferences.
RetrievedReference	A passage retrieved for a recommendation, with rank, score, and citation flag.	Refers to one KnowledgeBaseChunk.
StaffDecision	The reviewer's final decision (confirm, adjust, or manual) with reason and direction.	May amend an earlier decision (FR-48).
ClinicalNote	Free-text note added by a reviewer.	Written by one User.
AuditEntry	An append-only record of an action with before and after values.	Cannot be modified or deleted (FR-43).
KnowledgeBaseEntry	A curated reference entry for one presenting complaint with its source citation and status.	Composed of chunks and red-flag rules; approved by a User with the approval permission.
RedFlagRule	A red-flag indicator and its minimum VTL category.	Applied by the safety floor (FR-23).
KnowledgeBaseChunk	A passage of an entry and its embedding vector.	Target of RetrievedReferences.
KnowledgeBaseVersion	An immutable snapshot of the Active entries.	Referenced by every recommendation and evaluation run (FR-54).

CLASS	RESPONSIBILITY	KEY RELATIONSHIPS AND CONSTRAINTS
PresentingComplaint	An item of the fixed supported complaint list.	Shared by ExtractionResult and KnowledgeBaseEntry.
EvaluationSet / Vignette	A labeled test set of 50–100 vignettes with reference categories.	Stored separately from clinical cases (DR-06).
EvaluationRun / EvaluationResult	One execution of the pipeline over a set, with metrics and per-vignette outputs.	Uses one KnowledgeBaseVersion for reproducibility (NFR-23).

Model consistency. The classes of model (a) map to data stores D1 to D4 and those of model (b) to D5 and D6 (Section 6.1.3). The CaseStatus enumeration matches the triage case states in SRS Appendix B, the VTLCategory enumeration matches the five VTL categories in SRS Section 4.3, and the UserRole enumeration matches the roles in SRS Section 5.5.